Spectral Partitioning Residual Network With Spatial Attention Mechanism for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is one of the most important tasks in hyperspectral data analysis. Convolutional neural networks (CNN) have been introduced to HSI classification and achieved good performance. In this article, an effective and efficient CNN-based spectral partitioning residual network (SPRN) is proposed for HSI classification. The SPRN splits the input spectral bands into several nonoverlapping continuous subbands and uses cascaded parallel improved residual blocks to extract spectral–spatial features from these subbands, respectively. Finally, the features are fused and fed into a classifier. By equivalently using grouped convolutions, the spectral partition and feature extraction are embedded into an end-to-end network. Experimental results show that the proposed SPRN achieves state-of-the-art performance, meanwhile, with relatively fewer parameters and computational costs. Usually, the CNN takes a patch that contains continuous spatial information as the input and results in a class label of the center pixel. The large size of the input patch includes more spatial information, whereas also introduces interfering pixels that may lead to a degradation of classification accuracies. For that reason, we propose a novel spatial attention module named homogeneous pixel detection module (HPDM). The module alleviates the degradation of performance as the input patch size increases by capturing the homogeneous pixels in the input patch. The module can be integrated into any CNN-based HSI classification framework.**

***Index Terms*— Grouped convolutions, hyperspectral image (HSI) classification, spatial attention mechanism, spectral partition (SP).**

I. Introduction

SINCE hyperspectral images (HSIs) usually have hundreds of channels that are the continuous and narrow frequency bands of reflected radiation from a specific surface, HSIs have been widely applied in many areas [1]–[5]. This article mainly focuses on the classification task that aims at assigning every pixel in HSIs to a certain land-cover type. There are two major characteristics of HSIs that must be considered for HSI classification: 1) abundant spectral information that makes the accurate pixelwise classification possible and 2) strong spatial correlation that reflects the spatial distribution of different land types.

In the early studies of HSI classification, spectral information has been widely explored. Many classifiers, such as multinomial logistic regression (MLR) [6], neural networks [7], support vector machines (SVM) [8], decision trees [9], random forests [10], and sparse representation classifiers [11], [12], have been introduced to HSI classification, using one single pixel with all its bands as input. Nevertheless, large amounts of bands of hyperspectral data also lead to challenges. Due to the scarcity of labeled training samples, the high dimensionality of data may cause the Hughes phenomenon [13]. In that case, many dimension-reduction techniques have been proposed to extract features and reduce redundant information and noise from original spectral bands. Such techniques include the principal component analysis (PCA) [14], the linear discriminant analysis (LDA) [15], more recently, sparse representation [16]–[18], manifold learning [16], [17], stacked autoencoder (SAE) [19], and so on. However, the discriminative spectral information of HSIs may be lost when doing these dimension-reduction methods.

Spectral partition (SP) is an alternative strategy trying to solve the curse of dimensionality of hyperspectral data and avoid losing any useful information [20]–[24]. As the name suggests, SP splits the original spectral bands into several groups of nonoverlapping continuous subbands. After SP operation, feature extraction (FE) or classification is applied to each group, and final results are obtained by using a fusion strategy. The idea of SP comes from two facts: physical characteristics and statistical analysis. A spectral imaging system usually uses several spectrometers, and each produces signals in a certain range of wavelengths to generate a complete spectral signal. For example, the AVIRIS system [25] composes four different spectrometers, covering the wavelength ranges: 400–700, 700–1300, 1300–1900, and 1900–2500 nm, respectively. For a given pixel in HSIs, different spectrometers reflect different types of spectral features, and each feature type can be analyzed independently. From the statistical perspective, it is well known that hyperspectral data have the unique characteristic of grouped features. The correlation matrix of spectral bands illustrated in [21] shows high values in some continuous

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regions, which indicates that the bands of hyperspectral data have a strong correlation, especially with their adjacent bands. These continuous subbands share discriminative information about certain material properties. Therefore, SP is a natural and rational strategy that preserves the original bands and reduces the input dimensionality of a model as well. In this article, we take the advantages of the SP strategy in our classification framework.

Besides spectral information, more and more HSI classification frameworks have incorporated spatial information to make full use of hyperspectral data and achieved better results [26]. Techniques such as extended morphological profiles (EMPs) [27], the Markov random field (MRF) [28] or conditional random field (CRF) [29], multiple kernel learning [30], and superpixel segmentation [31] have been integrated to build a spectral–spatial classification model. More recently, the convolutional neural network (CNN), one of the popular deep learning models, has been introduced to HSI classification. Different from the traditional hand-crafted feature-based models, CNN is designed to extract high-level spatial features via cascading convolutional layers, which shows the potential for better performance. In [32], classification was performed by using a standard 2-D CNN. Song *et al.* [33] introduced residual learning [34] to build a very deep residual network and fused the outputs of different hierarchical layers. Zhu *et al.* [35] introduced deformable 2-D CNN to learn spatial information in an adaptive manner. Cheng *et al.* [36] used deep CNNs that are successful on ImageNet to extract deep features from HSIs and proposed metric learning method for feature fusion, feature learning, and classifier training. These methods applied PCA to reduce the input dimension. To fully use the abundant spectral information of HSIs, in [37], there existed two CNNs: one was a 1-D-CNN to extract spectral features along spectral dimension; the other used a 2-D-CNN to extract spatial features. The spectral features and the spatial features were concatenated and then fed into a classifier. In [38], a 3-D-CNN was proposed to extract spectral–spatial features jointly. Zhong *et al.* [39] and Wang *et al.* [40] designed spectral and spatial blocks to consecutively learn spectral–spatial features under the construction of 3-D-CNN. These 3-D-CNN classification frameworks get better performance and require fewer parameters than their 2-D counterparts, whereas they need higher computational costs.

In this article, we combine the SP strategy with 2-D-CNN to get better classification results and reduce the computational complexity. We propose the spectral partitioning residual network (SPRN): an effective and efficient HSI classification framework based on 2-D-CNN and residual learning together with the SP strategy. In SPRN, the input patch is partitioned into several groups with equal width along spectral dimension; then, they are fed into several parallel convolutional networks, one on each group, to get features. The features from parallel CNNs are fused in the final fusion layer. It should be noticed that we do not really construct many CNNs in our framework; instead, we use grouped convolutions to embed the SP and FE into one network. Thus, the SPRN is an easy-practiced end-to-end HSI classification framework. Moreover, we improve the residual block by adding an additional branch so that the parallel CNNs are not absolutely independent.

There are already some of the works that combine the SP strategy with deep learning methods. In [41], the spectral bands are partitioned before feeding into a deep belief network (DBN) together with a grouped weight-decay process designed as regularization. Under CNN architecture, after implementing an equal-width SP, BASSNet [24] applies several parallel networks, one on each group, to extract features. The outputs of these parallel networks are concatenated and fed into a set of full connected layers followed by a softmax layer. Zabalza *et al.* [42] use correlation matrix to guide the partition; Paul and Kumar [43] use the mutual information as the criterion, and they both apply SAE to extract features from each subband. However, these networks are too shallow to get better classification results, and these methods are relatively complicated to implement. SPRN is able to solve the shortcomings of these methods.

Based on the assumption that adjacent pixels are more likely to have the same labels, the input of a CNN-based HSI classification framework is a cube with size $p \times p \times B$ also called a patch formed by a given pixel and its neighbors, where the p is the patch size given in advance and the B is the number of spectral bands. After going through the CNN, a label is obtained and then assigned to the center pixel of the input patch. A CNN benefits from a large input patch size since it involves more spatial information; still, a large patch may contain interfering pixels whose labels are different from the center pixel of the patch, resulting in a decline in classification accuracies as the patch size increase. Thus, in order to reduce the influences of those interfering pixels, in other words, pay more attention to those pixels that similar to the center in a patch; spatial attention mechanism [44], [45] has been introduced to CNN-based HSI classification in some recent works. For example, Sun *et al.* [46] introduced the nonlocal attention block [44] to their spectral–spatial CNN to make the classification of the edge of land-cover more accurate. In [47], under the double-branch architecture, a spatial attention block proposed in [45] was introduced in the spatial branch to extract more discriminative spatial features. In [48], a spatial attention block was proposed to get better results. To further improve the classification accuracies and alleviate the degradation of performance when choosing a large input patch size, inspired by the previous works, a novel spatial attention module named homogeneous pixel detection module (HPDM) is proposed. The HPDM is able to capture the homogeneous pixels of a patch based on the spectral signature of the center pixel. The module can be integrated into the head of any CNN-based classification framework and trained jointly with the classification framework.

In summary, the contributions of this article are listed as follows.

1) A novel spectral partitioning residual network (SPRN) is constructed for HSI classification. The SP strategy is applied to partition the input patch along the spectral dimension, and several parallel 2-D-CNNs are constructed to extract spectral features, respectively. The residual blocks in SPRN are improved by adding an

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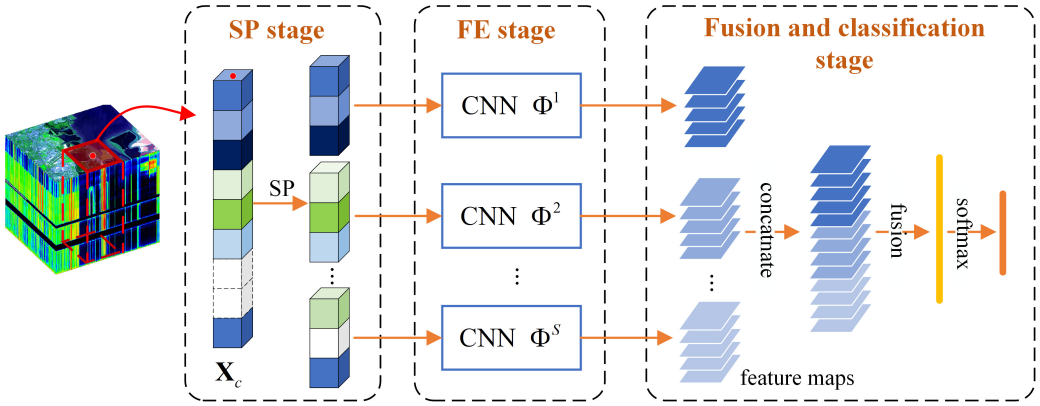


Fig. 1. HSI classification framework based on the SP strategy. The framework is mainly divided into three stages: the SP, the FE, and the fusion and classification.

additional branch so that the parallel CNNs are connected to each other. By using grouped convolution, the SPRN is equivalently applied in an easy-practiced way.

2) According to the experiments, SPRN is able to get better performance than HSI classification frameworks based on 2-D-CNN and more efficient than those based on 3-D-CNN.
3) A novel spatial attention block named HPDM is proposed to alleviate the degradation of performance when choosing a large patch size. The module can be integrated into the head of any CNN-based HSI classification framework.

The rest of this article is organized as follows. Section II reviews some related works about SP and CNN. Section III gives the details about the proposed SPRN. Section IV is the proposed spatial attention module HPDM. Section V shows the experimental results. In Section VI, we draw some conclusions.

II. Related Works

A. Spectral Partition

SP has been applied in previous HSI classification frameworks. In [20], the design of the imaging system was taken into account for partitioning the spectral bands. The work in [21] and [42] used spectral correlation coefficient matrix to guide the partition. Zhao *et al.* [22] partitioned the spectral bands under similar guidance and extracted the features from each group by the mean shift clustering. In [23], the bands were first divided into subbands with equal width; then, a fine partition strategy was proposed to make the former partition accurate. In [43], mutual information is used as the criterion to partition the spectrum.

Specifically, for an input patch, $\mathbf{X}_c \in \Re^{p\times p\times B}$, where p is the patch size and B is the number of spectral bands. SP is applied to split the input patch into S nonoverlap continuous subbands along the spectral dimension. Thus, the SP operation can be described as follows:

$$\left\{\mathbf{X}_c^{(0,a_1]}, \mathbf{X}_c^{(a_1,a_2]}, \ldots, \mathbf{X}_c^{(a_{S-1},\mathrm{B}]}\right\} = SP(\mathbf{X}_c) \tag{1}$$

where $\mathbf{X}_c^{(a_{j-1},a_j]} \in \Re^{p\times p\times(a_j-a_{j-1})}$ represents the subbands or subpatches of $\mathbf{X}_c$ from the spectral channel interval $(a_{j-1}, a_j]$ and obviously satisfies $\mathrm{X}_c^{(a_{j-1},a_j]} \cap \mathrm{X}_c^{(a_{j'-1},a_{j'}]} = \emptyset$ if $j \neq j'$. Function $SP(\cdot)$ decides how to partition the bands.

B. HSI Classification Framework Based on SP

An HSI classification framework based on SP is illustrated in Fig. 1. Let $\mathbf{X} \in \Re^{W\times H\times B}$ be the HSI data volume, where W and H are the image sizes and B is the number of spectral channels. Given a spectral signature $\mathbf{x}_c \in \Re^B$ from $\mathbf{X}$, the input patch $\mathbf{X}_c \in \Re^{p\times p\times B}$ is formed by $\mathbf{x}_c$ and its $p \times p$ neighbors. The proposed framework includes three stages: SP, FE, and fusion and classification. In the SP, $\mathbf{X}_c$ is partitioned into S subpatches along the spectral dimension. In the FE, S CNNs are constructed to extract the features of these subpatches, respectively. Finally, in the fusion and classification, the feature maps obtained by S CNNs are fused and fed into the classifier.

1) Spectral Partition: SP is applied to split the input patch $\mathbf{X}_c$ into S nonoverlap continuous subbands along the spectral dimension. Under our CNN architecture, we simply split the spectral bands with equal bandwidth b, where $b = B/S$. If B is not divisible by S, we pad the spectral bands into a new shape B', which means that the final values of the bands are copied to expand the spectral bands until they reach the new size B'. Then, we get

$$\left\{\mathrm{X}_c^{(0,b]}, \mathrm{X}_c^{(b,2b]}, \ldots, \mathrm{X}_c^{((S-1)b,B]}\right\} = SP(\mathrm{X}_c). \tag{2}$$

2) Feature Extraction: After the SP operation, S parallel CNNs are constructed to extract spectral–spatial features, respectively, from S subpatches

$$\mathbf{y}_c^k = \Phi^k\left(\mathbf{X}_c^{((k-1)b,kb]}\right), \quad (k = 1, 2, \ldots, S) \tag{3}$$

where vector $\mathbf{y}_c^k$ is the feature obtained by network $\Phi^k(\cdot)$. The parallel networks in our framework have the same CNN structure.

3) Feature Fusion and Classification: In the fusion stage, S features extracted by S parallel networks are concatenated, represented as

$$\mathbf{y}_c = \left[\mathbf{y}_c^1, \mathbf{y}_c^2, \ldots, \mathbf{y}_c^S\right]. \tag{4}$$

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Then, the concatenated features are summarized by a normal convolutional layer with kernel size 1×1 using ReLU as the activation function and a global average pooling (GAP) layer to reduce the spatial size of input patch into 1×1. Finally, we perform a softmax regression to get the final classification results. The outputs of the classifier are the conditional probabilities of the C output classes, which can be used for getting final results. Suppose that the conditional probabilities are $p = [p_1, p_2, \ldots, p_C]$; then

$$p_j = \frac{\exp\left(\theta_1^{(j)}\mathbf{z} + \theta_2^{(j)}\right)}{\sum_i^C \exp\left(\theta_1^{(j)}\mathbf{z} + \theta_2^{(j)}\right)}, \quad (j = 1, 2, \ldots, C) \tag{5}$$

where $\mathbf{z}$ is the input of the classifier, and θ_1 and θ_2 are the parameters. We train the framework by minimizing cross-entropy loss function. Given the training set $\{\mathbf{X}_i, \mathbf{t}_i\}_{i=1}^N$, the loss function is given by

$$L = -\sum_{i=1}^{N}\sum_{j=1}^{C} \mathbf{t}_j^{(i)} \log p_j^{(i)}. \tag{6}$$

C. CNN and Residual Learning

A 2-D-CNN is constructed by stacking multiple 2-D convolutional layers. Let $\mathbf{O}^{l-1} \in R^{p\times p\times n^{l-1}}$ be the output or the feature maps of the $l-1$th convolutional layer, where $p \times p$ is the spatial size and n^{l-1} represents the number of the feature maps. Suppose that the lth convolutional layer has n^l filters; then, the output of the convolutional layer can be formulated as

$$\mathbf{O}_j^l = \sigma\left(\sum_{i=1}^{n^{l-1}} \mathbf{O}_i^{l-1} * \mathbf{w}_{ij}^l + b_j^l\right), \quad j = 1, 2, \ldots, n^l \tag{7}$$

where $*$ represents the spatial convolution operation, $\mathbf{O}_i^{l-1}$ represents the ith feature map of $\mathbf{O}^{l-1}$, and $\mathbf{w}_{ij}^l$ and b_j^l are weights and bias. $\sigma(\cdot)$ is the activation function; here, we use rectified linear unit (ReLU), which is

$$\sigma(x) = \max(0, x). \tag{8}$$

The batch normalization (BN) layer [49] is connected after every convolutional layer, which is able to speed up the training process. A BN layer can be formulated as follows:

$$\widehat{\mathbf{O}_j^l} = \frac{\mathbf{O}_j^l - \mu_B}{\sigma_B} \tag{9}$$

$$\mathbf{O'}_j^l = \gamma\,\widehat{\mathbf{O}_j^l} + \beta \tag{10}$$

where $\mathbf{O}_j^l$ is the input, $\mathbf{O'}_j^l$ is the output of the BN layer, and μ_B and σ_B are the expectation and variance of one batch data. γ and β are the learnable parameters.

As the CNN goes deeper, the training of the network becomes harder, leading to the degradation of the performance. Suppose that $F(\mathbf{O})$ is the function that should be learned by a CNN or a convolution layer, which maps the input $\mathbf{O}$ into its feature space. Since the residual mapping is easier to optimized than the original mapping [34], the mapping function $F(\mathbf{O})$ is replaced with the residual function $G(\mathbf{O}) = F(\mathbf{O}) - \mathbf{O}$ by adding additional skip connections between different convolution layers. This is also well known as the residual learning [34] that is able to alleviate the degradation of performance in a deep network. Fig. 2 gives the basic structure of a residual block. The residual function $G(\mathbf{O})$ in our CNNs is built by two convolutional layers as a standard residual block does, together with BN after each convolution before activation to speed up the convergence of the CNNs. Thus, we have

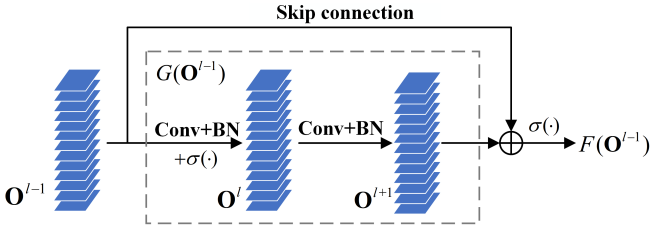


Fig. 2. Residual block.

$$F(\mathbf{O}^{l-1}) = \sigma(\mathbf{O}^{l+1} + \mathbf{O}^{l-1}) \tag{11}$$

$$\mathbf{O}^{l+1} = f_{\mathrm{BN}}(\mathbf{w}_2 * \mathbf{O}^l + \mathbf{b}_2) \tag{12}$$

$$\mathbf{O}^l = \sigma(f_{\mathrm{BN}}(\mathbf{w}_1 * \mathbf{O}^{l-1} + \mathbf{b}_1)) \tag{13}$$

where $\mathbf{O}^l$ means the output of the lth layer, $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{b}_1, \mathbf{b}_2\}$ are weights and bias, and $\sigma(\cdot)$ represents the activation function.

III. Proposed SPRN for HSI Classification

A. Spectral Partitioning Residual Network

This section develops SPRN for HSI classification under the framework that we discussed in Section II-B. The parallel CNNs in the FE stage are supposed to respectively extract deep features from different groups of subbands that are obtained by the SP operation. All the parallel networks share the same structure but have different parameters. In SPRN, we do not really build these parallel CNNs; instead, we use grouped convolutions to make an equivalent implement. Besides, we further develop the residual blocks so that each CNN is able to make a communication with others in their internal layers. The improved residual blocks include two branches: the local branch and the global branch. The details about the grouped convolution equivalent and the improved residual block are discussed in the following.

1) Grouped Convolutions Equivalent: Fig. 3(a) shows the structure of parallel 2-D-CNN residual blocks (only convolutional layers are showed), each of which belongs to one subpatch. Fig. 3(b) and (c) shows the equivalent implements of Fig. 3(a). Fig. 3(b) concatenates the S features so that the addition can be applied in one vector. Fig. 3(c) shows that the S convolution layers in the same depth of the parallel CNNs can be replaced by one 2-D grouped convolutional layer [50]; the number of the groups equals S. Moreover, if we apply the grouped convolution to the input patch, it will split the input channels into S groups with equal channel width, which is exactly the same effect as the SP operation does. Thus, there is no need to build many parallel networks for implementation. We replace all the convolutional layers with the grouped ones.

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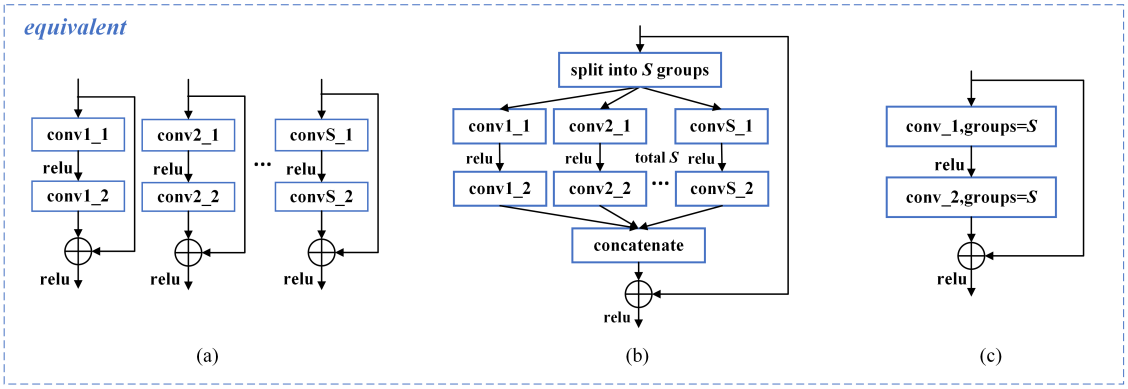


Fig. 3. Equivalent building blocks of parallel residual blocks. (a) Original parallel residual blocks. (b) Equivalent implement by concatenating tensors. (c) Equivalent implement by using grouped convolutions with the group number equal to S.

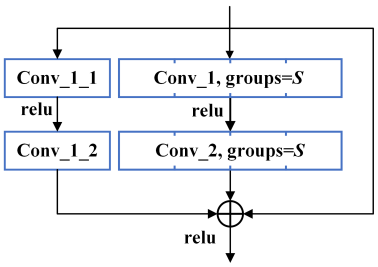


Fig. 4. Improved residual block includes two branches: local branch and global branch. The local branch is built by two grouped convolutions layers, while the global branch is built by two normal convolutional layers.

2) Local Branch and Global Branch: We use the grouped convolutions to construct the parallel CNNs. However, these parallel CNNs extract features from certain subbands independently, which lacks the global information of the whole bands. To overcome these problems, we improve the residual blocks. As illustrated in Fig. 4 (only convolutional layers are showed), the improved residual block includes two branches: local branch and global branch. The local branch is constructed by two grouped convolutional layers that are used for building S parallel CNNs. The grouped convolutions extract local spectral features from a certain channel interval. The global branch is built by two normal convolutions that are used to extract global spectral features from the whole input channels. The local spectral features and the global spectral features are fused by addition.

Finally, the complete and final architecture of SPRN is illustrated in Fig. 5 (only convolutional layers are showed). The input size of the SPRN is $p \times p \times B$, where the p is the patch size and the B is the band size. The first grouped convolutional layer (GC1) splits the input bands into S groups. Then, two cascaded improved residual blocks are connected to extract spectral features. GC2–GC5 are grouped convolutional layers, and C1–C4 are normal convolutional layers. A normal convolutional layer (C5) is applied to fuse the spectral features, and the GAP layer is used for fusing the spatial features. We pad the feature maps in the SPRN to keep the spatial size; thus, the patch size $p \times p$ does not change until the GAP layer. Finally, the softmax layer gives the final output.

Algorithm 1 Details of the HPDM

Input: Input data patch $\mathbf{X}_c \in \Re^{p\times p\times B}$

Output: Revised data patch $\mathbf{X}'_c \in \Re^{p\times p\times B}$

1: Apply 1×1 convolution with B/2 filters to input patch together with BN and sigmoid activation to obtain $\mathbf{Q}_c \in \Re^{p\times p\times B/2}$

2: **for** $\mathbf{q}_c, \mathbf{q}_{i,j} \in \Re^B$ in $\mathbf{Q}_c$($\mathbf{q}_c$ is the center pixel of $\mathbf{Q}_c$, $i, j = 1, 2, \ldots, p$) **do**

3: Calculate the cosine similarity to get the attention weight: $m_{i,j} = \frac{\mathbf{q}_{i,j}^T \cdot \mathbf{q}_c}{\|\mathbf{q}_{i,j}\| \cdot \|\mathbf{q}_c\|}$

4: **end for**

5: Get attention mask $\mathbf{M}_c \in \Re^{p\times p}$

6: Get revised patch by calculating the spatialwise product $\mathbf{X}'_c = \mathbf{M}_c \odot \mathbf{X}_c$

We use 1×1 kernel size in all convolutional layers. Suppose that GC1–GC5, C2, and C4 have c_g filters, C1 and C3 have c filters, and C5 has c_d filters. The parameters and FLOPs of SPRN are listed in Table I. The HPDM will be discussed in the next section.

IV. Homogeneous Pixel Detection Module

In this section, we propose a novel spatial attention module named HPDM to capture the homogeneous pixels in a patch. More specifically, the module generates an attention weight mask that can measure the importance of every pixel in a patch. Considering that the center pixel plays the leading role in a patch, we get the weight mask by calculating the similarity between the center pixel and the others. Homogeneous pixels whose spectral signatures are similar to the center pixel are more likely to gain higher scores while the interfering pixels gain lower.

There are many functions that can be used to measure the similarity. In this article, we use the cosine similarity as the criteria, which is also called the spectral angle in the HSI theme. Specifically, given the input patch $\mathbf{X}_c$, the attention weight masks $\mathbf{M}_c \in \Re^{p\times p}$ are calculated by

$$m_{i,j} = \frac{\varphi(\mathbf{x}_{i,j})^T \cdot \varphi(\mathbf{x}_c)}{\|\varphi(\mathbf{x}_{i,j})\| \cdot \|\varphi(\mathbf{x}_c)\|}, \quad (i, j = 1, 2, \ldots, p) \tag{14}$$

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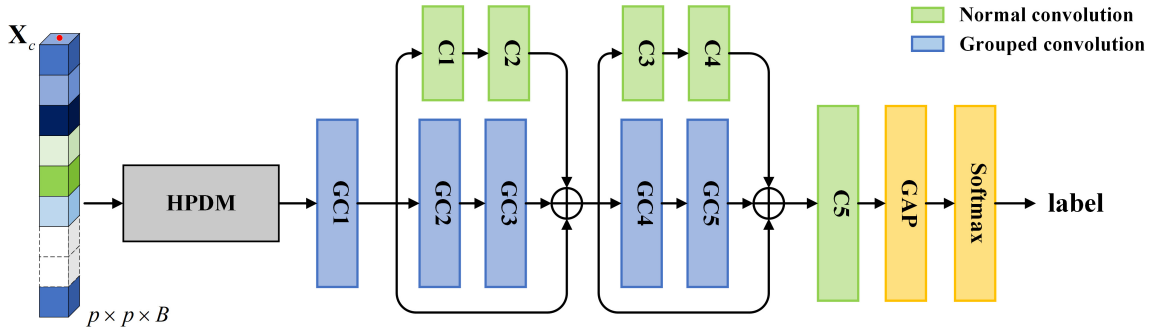


Fig. 5. Architecture of SPRN. The blue blocks use the grouped convolution, while the green blocks use the normal convolution.

TABLE I
PARAMETERS AND FLOPS OF SPRN

	GC1	GC2;C1	GC3;C2	GC4;C3	GC5;C4	C5	Softmax
Params.	$Bc_g/g + c_g$	$c_g^2/g + c_g;\ c_g c + c$	$c_g^2/g + c_g;\ cc_g + c_g$	$c_g^2/g + c_g;\ c_g c + c$	$c_g^2/g + c_g;\ cc_g + c_g$	$c_g c_d + c_d$	$c_d C + C$
FLOPs	$Bc_g p^2/g$	$c_g^2 p^2/g;\ c_g c p^2$	$c_g^2 p^2/g;\ cc_g p^2$	$c_g^2 p^2/g;\ c_g c p^2$	$c_g^2 p^2/g;\ cc_g p^2$	$c_g c_d p^2$	$c_d C$
Total							
Params.	$c_g \cdot (B/g + 4c_g/g + 4c + c_d + 5) + 2c + c_d + c_d C + C$						
FLOPs	$c_g \cdot (B/g + 4c_g/g + 4c + c_d) \cdot p^2 + c_d C$						

where $\mathbf{x}_{i,j} \in \Re^B$ is the spectral signature of the pixel in the ith row and the jth column of the patch $\mathbf{X}_c$, and vector $\mathbf{x}_c \in \Re^B$ is the center one. $m_{i,j}$ is the weight in the ith row and the jth column of the attention weight mask $\mathbf{M}_c$. $\|\cdot\|$ refers the two-norm operator, and $\varphi(\cdot)$ is a function that maps the original signatures into their feature space and reduces high channel dimensions. Then, the weight mask is multiplied with the original input patch to obtain the revised input $\mathbf{X}'_c$, which is

$$\mathbf{X}'_c = \mathbf{M}_c \odot \mathbf{X}_c \tag{15}$$

where $\odot$ is a spatialwise product. For the function $\varphi(\cdot)$, we apply a 1×1 spatial convolutional layer to the input patch $\mathbf{X}_c$ to reduce the original dimension B to $B/2$, which is

$$\varphi(\mathbf{X}_c) = \sigma(\text{conv}(\mathbf{X}_c)) \tag{16}$$

where conv$(\cdot)$ represents the spatial convolution with kernel size 1×1, and $\sigma(\cdot)$ is the activation function; here, we use sigmoid function to limit the feature values in $(0, 1)$. The sigmoid function is represented by

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{17}$$

It can be easily found that the attention weights that we got satisfy $[0, 1]$ due to the nature of cosine similarity. The complete implement of HPDM is illustrated in Fig. 6. We first apply a 1×1 convolution to reduce the input channels to $B/2$. Then, we use the sigmoid function to limits the value in $(0, 1)$. The cosine similarity is calculated between the center pixel and its neighbors to get the spatial mask $\mathbf{M}_c$. Finally, the mask is multiplied with the original input patch to produce a revised input patch. Parameters of the convolutional layer in HPDM are initialized to 0 and can be trained together with other stages in the framework.

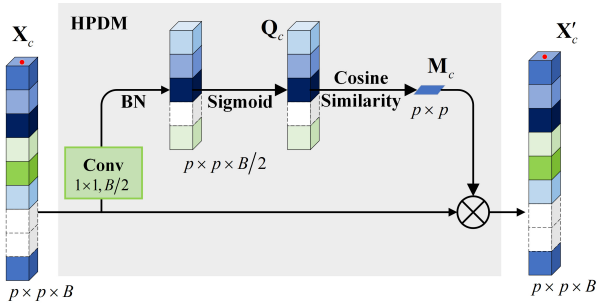


Fig. 6. Proposed HPDM.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we design several experiments to verify our proposed SPRN and the HPDM on three popular datasets[1]: Indian Pines, Salinas, and Pavia University.

A. Datasets

The Indian Pines scene was gathered by the 224-band Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the Indian Pines test site in Northwestern Indiana, covering the wavelength range from 0.4 to 2.5 um. The image size is 145×145 and contains 16 interested classes. The number of bands is reduced to 200 by removing several water absorption bands.

The Salinas scene was also collected by AVIRIS over Salinas Valley, California. The image size is 512×217 and contains 16 classes; 20 water absorption bands are discarded in experiments.

[1]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_51Scenes

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TABLE II
NUMBERS OF TRAINING AND TESTING SAMPLES ON THE INDIAN PINES DATASET

No.	Classes	Train./Val.	Test.
1	Alfalfa	2/2	42
2	Corn-notill	71/71	1286
3	Corn-mintill	41/41	748
4	Corn	12/12	213
5	Grass-pasture	24/24	435
6	Grass-trees	37/37	656
7	Grass-pasture-mowed	1/1	26
8	Hay-windrowed	24/24	430
9	Oats	1/1	18
10	Soybean-notill	48/48	876
11	Soybean-mintill	123/123	2209
12	Soybean-clean	30/30	533
13	Wheat	10/10	185
14	Woods	63/63	1139
15	Buildings-Grass-Trees-Drives	19/19	348
16	Stone-Steel-Towers	5/5	83

The Pavia University scene was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) over Pavia, Northern Italy. The number of spectral bands is 103, covering the wavelength range from 0.38 to 0.86 um. The image size is 610 $\times$ 340, and 9 classes are considered for experiments.

The data values are normalized to the range [0, 1]. Also, we apply the data augment to the input patches. Specifically, we first flip the patch horizontally or vertically; then, the patch is randomly rotated 90°, 180°, or 270°.

B. Experimental Settings

For a fair comparison, all the HSI classification frameworks are implemented on Pytorch. We set the batch size to 64 and use the Adam [51] optimizer to update the training parameters in all frameworks. The epoch of training process is fixed to 300. The initial learning rate is set to 0.001; at the epoch of 100, it will multiply by a factor of 0.1, and at the epoch of 250, it will multiply by a factor of 0.01. For the Indian Pines dataset, 5% labeled samples are used for training, 5% are used for validation, and 90% are used for test. For Salinas and Pavia University, 0.5%, 0.5%, and 99% of labeled samples are selected for training, validation, and testing, respectively (see Tables II–IV). The best models on the validation set are evaluated on the testing set. We randomly split the labeled samples ten times, and all the results are the average of ten runs. Overall accuracy (OA), average accuracy (AA), and the kappa coefficient are adopted to measure the performance of different classification methods. OA is the ratio of the correct labels to the total number of labels, AA is the average of the accuracies of all categories, and kappa coefficient measures the consistence of predicted labels and ground truth. Suppose that $M \in \Re^{C\times C}$ is the confusion matrix of classification results. The values of OA, AA, and kappa can be calculated by the following formulas:

$$\mathrm{OA} = \frac{1}{N}\sum_{i}^{C} M_{ii} \tag{18}$$

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES ON THE SALINAS DATASET

No.	Classes	Train./Val.	Test.
1	Brocoli_green_weeds_1	10/10	1989
2	Brocoli_green_weeds_2	19/19	3688
3	Fallow	10/10	1956
4	Fallow_rough_plow	7/7	1380
5	Fallow_smooth	13/13	2652
6	Stubble	20/20	3919
7	Celery	18/18	3543
8	Grapes_untrained	56/56	11159
9	Soil_vinyard_develop	31/31	6141
10	Corn_senesced_green_weeds	16/16	3246
11	Lettuce_romaine_4wk	5/5	1058
12	Lettuce_romaine_5wk	10/10	1907
13	Lettuce_romaine_6wk	5/5	906
14	Lettuce_romaine_7wk	5/5	1060
15	Vinyard_untrained	36/36	7196
16	Vinyard_vertical_trellis	9/9	1789

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES ON THE PAVIA UNIVERSITY DATASET

No.	Classes	Train./Val.	Test.
1	Asphalt	33/33	6565
2	Meadows	93/93	18463
3	Gravel	10/10	2079
4	Trees	15/15	3034
5	Painted metal sheets	7/7	1331
6	Bare Soil	25/25	4979
7	Bitumen	7/7	1316
8	Self-Blocking Bricks	18/18	3646
9	Shadows	5/5	937

$$\mathrm{AA} = \frac{1}{C}\sum_{i}^{C}\frac{M_{ii}}{M_{+i}} \tag{19}$$

$$\mathrm{Kappa} = \frac{N\sum_{i}^{C} M_{ii} - \sum_{i}^{C} M_{i+}M_{+i}}{N^2 - \sum_{i}^{C} M_{i+}M_{+i}} \tag{20}$$

where M_{i+} represents the sum of the column of M and M_{+i} is the sum of the row of the M. All frameworks are trained on a NVIDIA GTX 1080Ti GPU. The code is available online.[2]

C. Number of Groups in SPRN

The group number of grouped convolutions in SPRN depends on the number of subbands split by SP operation and the number of parallel CNNs that extract features from these subbands. Therefore, it is an important parameter that needs to be considered. In this section, we explore the performance of SPRN with a different number of groups to find the best settings for different datasets. We fix the capacity of SPRN about 180k parameters and change the number of groups and filters in grouped convolutional layers. Fig. 7 shows the OA of the experiment. From Fig. 7, it can be observed that the

[2]https://github.com/shangsw/HPDM-SPRN

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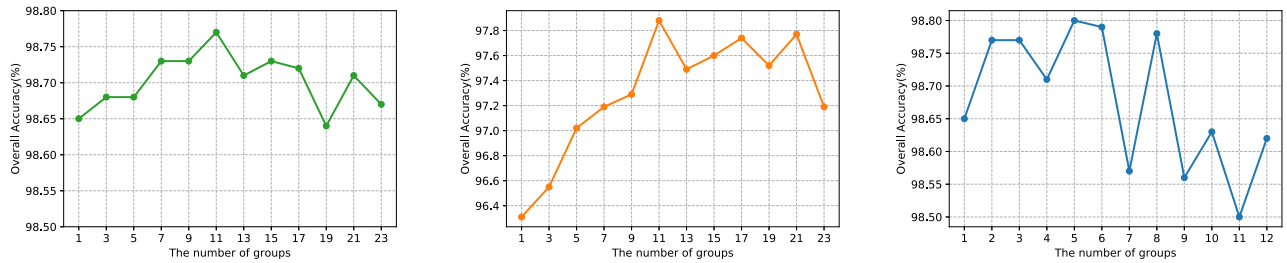


Fig. 7. Overall accuracies of SPRN with different group numbers on (a) Indian Pines, (b) Salinas, and (c) Pavia University.

TABLE V
ARCHITECTURE OF SPRN ON INDIAN PINES, SALINAS, AND PAVIA UNIVERSITY DATASETS

Layer name	Filter numbers		
	Indian Pines	Salinas	Pavia University
GC1	407,g=11	407,g=11	320,g=5
GC2;C1	407,g=11;32	407,g=11;32	320,g=5;32
GC3;C2	407,g=11;407	407,g=11;407	320,g=5;320
GC4;C3	407,g=11;32	407,g=11;32	320,g=5;32
GC5;C4	407,g=11;407	407,g=11;407	320,g=5;320
C5	128	128	128

TABLE VI
CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET

	DFFN	DHCNet	SSRN	FDSSC	BASSNet	SPRN	HPDM-SPRN
OA	93.55 ±0.70	94.77 ±0.56	97.01 ±0.48	96.85 ±0.36	93.80 ±1.07	**98.29 ±0.43**	98.23 ±0.47
AA	90.45 ±2.37	91.93 ±1.92	95.55 ±0.82	95.62 ±1.38	91.48 ±2.40	**97.85 ±0.61**	97.84 ±0.75
$\kappa \times 100$	92.65 ±0.80	94.04 ±0.64	96.59 ±0.55	96.41 ±0.40	92.92 ±1.22	**98.05 ±0.49**	97.98 ±0.54
1	70.49	74.15	89.02	89.51	73.17	96.59	**97.56**
2	93.05	94.06	97.29	96.73	92.47	**98.98**	98.57
3	92.24	93.84	95.50	95.74	88.67	**97.24**	97.23
4	90.14	91.74	97.37	95.07	89.15	**98.73**	98.36
5	92.76	92.62	94.21	93.33	93.61	**95.49**	95.49
6	97.52	97.76	98.98	98.78	93.13	**99.54**	99.51
7	86.00	84.80	91.60	99.60	91.20	**100.00**	100.00
8	99.53	99.40	99.84	99.93	99.81	99.98	**100.00**
9	74.44	79.44	87.78	82.22	78.33	**94.44**	94.44
10	88.25	90.43	93.79	94.70	90.64	**96.90**	96.79
11	94.70	95.73	97.34	97.08	94.92	98.15	**98.24**
12	89.91	90.94	97.45	96.29	86.70	**98.82**	98.71
13	97.14	99.46	98.97	99.19	**99.62**	98.97	99.24
14	97.05	98.76	99.44	99.23	98.97	**99.72**	99.70
15	88.56	91.15	93.29	94.76	89.34	**96.08**	96.08
16	95.36	96.67	96.90	97.74	**97.86**	95.95	95.48

best number of groups varies from different datasets. For the Indian Pines and Salinas datasets, the best group number is 11; for the Pavia University dataset, the best OA appears when the group number is equal to 5. When the group number is 1, the grouped convolutions degenerate into the normal convolutions, which means that we do not apply the SP operation and use only one simple CNN to extract features from the whole bands. Thus, we can draw the conclusion that the SP operation is effective. We also notice that the SP does not always give better performance. This is because the equal width SP may destroy the wavelength ranges that include discriminative information. We set the best group numbers in our following experiments.

D. Comparison With Other CNN-Based Methods

In this section, we compare our SPRN and HPDM-SPRN with other CNN-based HSI classification frameworks. Specifically, we compare our framework with DFFN [33], DHCNet [35], SSRN [39], FDSSC [40], and BASSNet [24]. DFFN and DHCNet use 2-D-CNN; SSRN and FDSSC are based on 3-D-CNN. BASSNet is a CNN-based framework applying the SP operation. To make a fair comparison, the spatial patch size is set to 7×7 for all methods and datasets. We remove the max-pooling layers and set the convolution stride to 1 in DFFN and DHCNet so that they can handle such a small patch size. The detailed architectures of SPRN on three datasets are shown in Table V. We take the Indian Pines dataset for example. All grouped convolution layers have 407 filters with 11 groups. In other words, we partitioned the spectrum into 11 groups and use 11 CNNs to extract features separately, where each CNN includes 37 filters in its layers. For normal convolutional layers, C1 and C3 have 32 filters, while C2 and C4 have 407 filters, and C5 has 128 filters. All convolutional layers use 1×1 kernel size. We choose these numbers of filters to keep the network about 180k parameters. The classification results of different methods can be found in Tables VI–VIII.

As the results illustrate, our SPRN and HPDM-SPRN outperform the other methods on the three datasets. Compared with DFFN and DHCNet that are based on 2-D-CNN, the results obtained by SPRN and HPDM-SPRN are much better. DFFN is also a residual network, which indicates that the SP strategy is helpful for performance improvement. As for those based on 3-D-CNN, on the Indian Pines dataset, SPRN achieves 1.28% higher than SSRN and 1.44% higher than FDSSC in terms of OA and obtains better improvement on some categories with fewer training samples, such as alfalfa and oats. On the Salinas dataset, SPRN performs much better than others and exceeds SSRN and FDSSC by 2.86% and 3.61% respectively, and shows advantages in most of the

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TABLE VII
CLASSIFICATION RESULTS ON THE SALINAS DATASET

	DFFN	DHCNet	SSRN	FDSSC	BASSNet	SPRN	HPDM-SPRN
OA	93.05 ±0.60	93.02 ±0.51	94.60 ±0.71	93.85 ±0.57	92.72 ±0.79	**97.46 ±0.49**	97.43 ±0.70
AA	94.88 ±0.77	94.51 ±0.79	96.17 ±0.88	95.90 ±0.72	94.51 ±0.65	98.45 ±0.28	**98.49 ±0.39**
$\kappa \times 100$	92.27 ±0.67	92.23 ±0.57	93.99 ±0.79	93.16 ±0.63	91.91 ±0.87	**97.17 ±0.55**	97.14 ±0.78
1	97.41	96.26	98.67	99.67	99.51	**99.99**	99.99
2	99.08	98.78	100.00	99.98	99.51	**100.00**	100.00
3	90.02	89.67	93.40	92.41	93.05	99.19	**99.95**
4	96.82	96.15	98.98	98.16	96.67	**99.06**	98.70
5	94.63	94.96	96.08	97.00	90.41	**98.95**	98.91
6	99.60	99.73	99.91	99.95	99.73	**100.00**	99.97
7	99.63	99.56	99.95	99.79	99.40	**99.97**	99.97
8	85.12	87.64	89.19	87.59	83.89	**95.07**	93.71
9	98.73	99.27	99.56	99.87	99.89	**100.00**	100.00
10	90.81	89.80	93.91	92.82	90.69	**97.36**	97.08
11	88.52	88.52	94.34	92.41	84.28	**98.61**	98.39
12	99.50	99.19	**100.00**	99.99	98.93	99.97	99.96
13	97.49	98.50	97.32	**98.79**	97.36	98.00	97.97
14	98.04	97.44	96.93	97.93	97.60	**99.32**	99.05
15	86.77	84.20	86.69	83.45	86.24	91.79	**93.31**
16	93.98	94.45	93.73	94.61	93.06	**99.85**	98.67

TABLE VIII
CLASSIFICATION RESULTS ON THE PAVIA UNIVERSITY DATASET

	DFFN	DHCNet	SSRN	FDSSC	BASSNet	SPRN	HPDM-SPRN
OA	93.52 ±1.10	94.43 ±1.14	97.65 ±0.80	96.86 ±0.77	92.21 ±0.99	97.92 ±0.57	**97.99 ±0.53**
AA	90.21 ±1.95	91.32 ±2.11	96.50 ±1.24	94.98 ±1.43	87.58 ±1.62	**96.75 ±1.05**	96.60 ±1.16
$\kappa \times 100$	91.38 ±1.47	92.61 ±1.53	96.88 ±1.06	95.83 ±1.02	89.63 ±1.32	97.23 ±0.75	**97.33 ±0.70**
1	93.31	93.59	**99.38**	97.40	92.14	98.56	98.48
2	98.60	98.65	99.17	99.72	98.31	99.84	**99.91**
3	69.55	68.43	85.53	76.96	58.88	**87.11**	85.35
4	94.22	93.87	94.67	93.83	94.66	95.36	**95.40**
5	99.51	99.55	**99.99**	99.86	99.45	99.89	99.88
6	91.04	94.95	**98.77**	97.94	89.08	97.28	97.82
7	81.59	84.91	98.18	97.78	71.06	98.82	**98.91**
8	85.41	89.80	93.14	91.68	85.95	94.82	**96.13**
9	98.63	98.26	**99.69**	99.66	98.67	99.11	97.51

categories (13 out of 16 categories). On the Pavia University dataset, although the results obtained by SPRN are very close to SSRN, SPRN still has more competitive results with lower deviations. By combining the SP strategy with 2-D-CNN, the classification performance of SPRN is comparable to that based on 3-D-CNN and even much better in some datasets. In fact, SP reduces the input dimensionality of each CNN in the FE stage, and the spectral features of the whole bands are obtained by a fusion operation. Therefore, no spectral information is lost, and higher performance is achieved. BASSNet is also a 2-D-CNN HSI classification framework applied with the SP strategy; however, it does not provide satisfactory results.

TABLE IX
TRAINING TIMES OF DIFFERENT METHODS ON THREE DATASETS

		IP	Salinas	PU
DFFN	Params.	423k	423k	506k
	Train.(s)	135	87	72
	Test.(s)	8.6	17	30
DHCNet	Params.	1.26M	1.31M	1.22M
	Train.(s)	111	69	54
	Test.(s)	7.8	47	45
SSRN	Params.	364k	370k	217k
	Train.(s)	141	78	39
	Test.(s)	3.1	21	26
FDSSC	Params.	2.39M	2.44M	1.23M
	Train.(s)	336	189	90
	Test.(s)	9.0	44	47
BASSNet	Params.	158k	162k	62k
	Train.(s)	129	102	45
	Test.(s)	3.8	30	20
SPRN	Params.	183k	185k	179k
	Train.(s)	87	51	36
	Test.(s)	1.7	7.6	11
HPDM-SPRN	Params.	205k	206k	185k
	Train.(s)	105	63	39
	Test.(s)	2.3	12	12

This is because the network is built too shallow, and the parallel CNNs are independent, so the features are not extracted sufficiently. Deep structure and the improved residual block of SPRN give a great improvement in performance. We also notice that the superiority of HPDM-SPRN in comparison to SPRN is not significant; SPRN and HPDM-SPRN get nearly the same results. Because the patch size in this experiment is set to 7×7, there are few heterogeneous or interfering pixels in such a relatively small patch. In our later experiments, we increase the input patch size to clearly show how HPDM works. We choose one of ten experiments to visualize the predicted samples of different methods. The visualizations are illustrated in Figs. 8–10, and the training samples are included. On the Indian Pines dataset, SPRN shows obvious superiority in keeping the edge. On the Salinas dataset, the classification maps of SPRN and HPDM-SPRN have fewer noises. On the Pavia University dataset, the results generated by SPRN and HPDM-SPRN are more accurate than others.

Table IX gives the parameters and training times of these methods. By combining the SP strategy with 2-D-CNN, SPRN is faster than those based on 3-D-CNN (SSRN and FDSSC), meanwhile, with relatively fewer parameters than other 2-D-CNN methods. We also change the percentage of training samples, and the results can be found in Fig. 11. As the training samples increase, the performance gap between different methods is gradually narrowing. Our SPRN and HPDM-SPRN still achieve good results, which shows the robustness of our framework.

E. Impact of the Convolution Kernel Size

The kernel size of convolutional layers influences the performance. In this section, we change the convolution kernel size of SPRN on three datasets to explore the effects.

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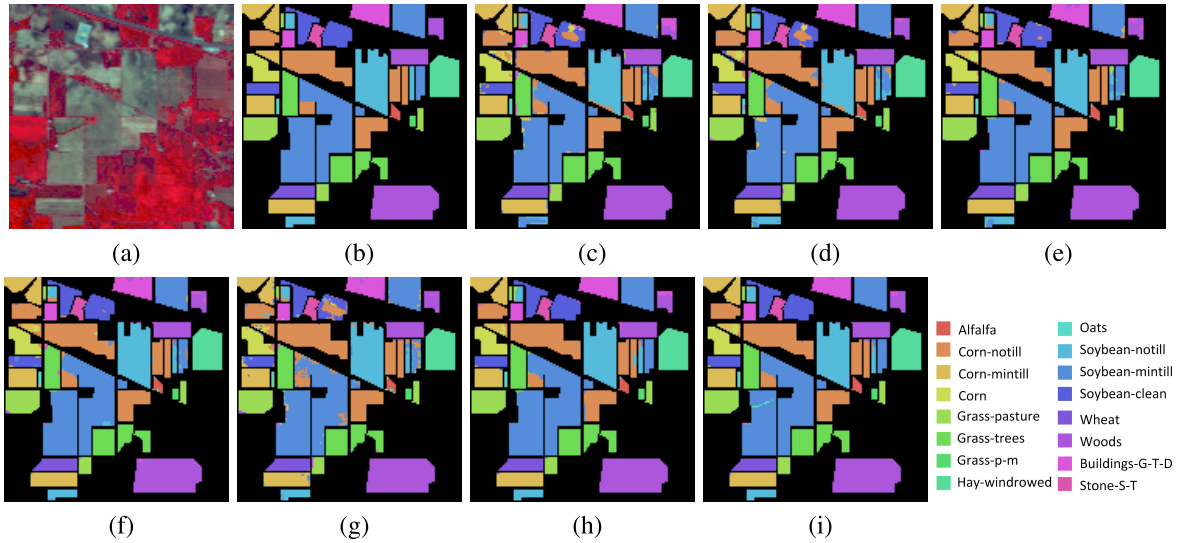


Fig. 8. Classification maps on the Indian Pines dataset. (a) False RGB image. (b) Ground truth. (c) DFFN. (d) DHCNet. (e) SSRN. (f) FDSSC. (g) BASSNet. (h) SPRN. (i) HPDM-SPRN.

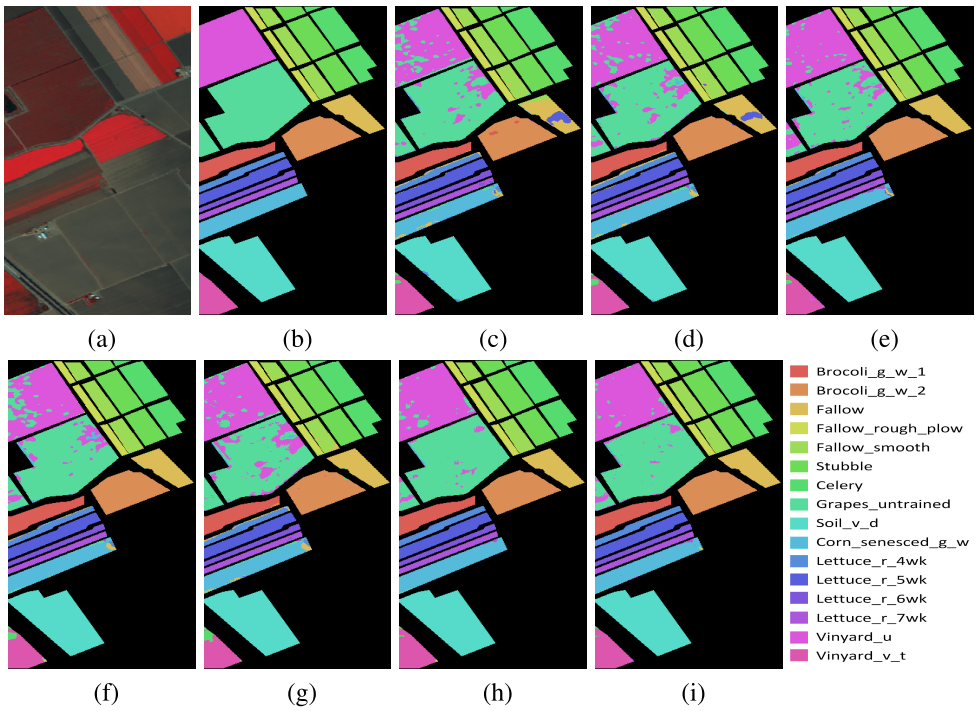


Fig. 9. Classification maps on the Salinas dataset. (a) False RGB image. (b) Ground truth. (c) DFFN. (d) DHCNet. (e) SSRN. (f) FDSSC. (g) BASSNet. (h) SPRN. (i) HPDM-SPRN.

The patch size is fixed to 7×7, and the numbers of training samples and testing samples are listed in Tables II–IV. The results are shown in Table X. As the kernel size increases, the OAs of classification decrease. In fact, for a small patch size, the spatial information of the input data patch is not so complicated. The low-frequency features obtained by GAP are enough for classification, and a large kernel size may overfit the model.

TABLE X
OA(%) OF SPRN WITH DIFFERENT CONVOLUTION KERNEL SIZES

	1×1	3×3	5×5	7×7
Indian Pines	98.14	96.74	95.19	94.15
Salinas	97.51	95.08	94.31	94.31
Pavia University	97.92	96.39	94.89	94.01

F. Impact of the Input Patch Size

The input patch size influences the classification. A larger patch size of the input patch usually contains more local spatial information. However, more interfering pixels are also included, which have a bad effect on FE and, finally, lead to a degradation of the model on classification performance. In this

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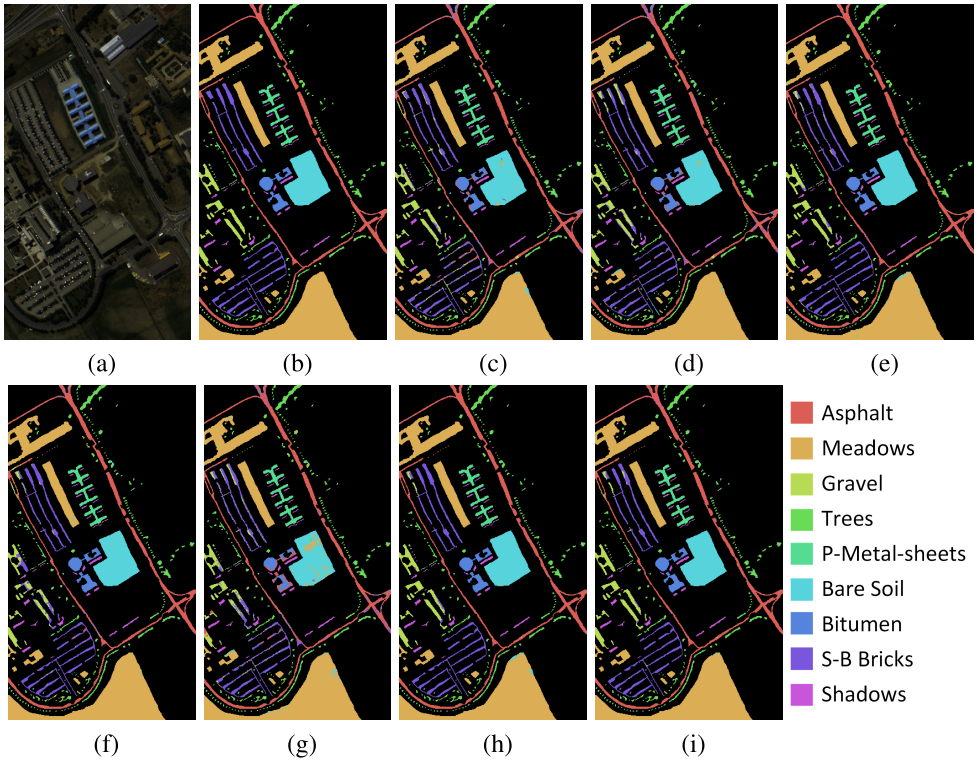


Fig. 10. Classification maps on the Pavia University dataset. (a) False RGB image. (b) Ground truth. (c) DFFN. (d) DHCNet. (e) SSRN. (f) FDSSC. (g) BASSNet. (h) SPRN. (i) HPDM-SPRN.

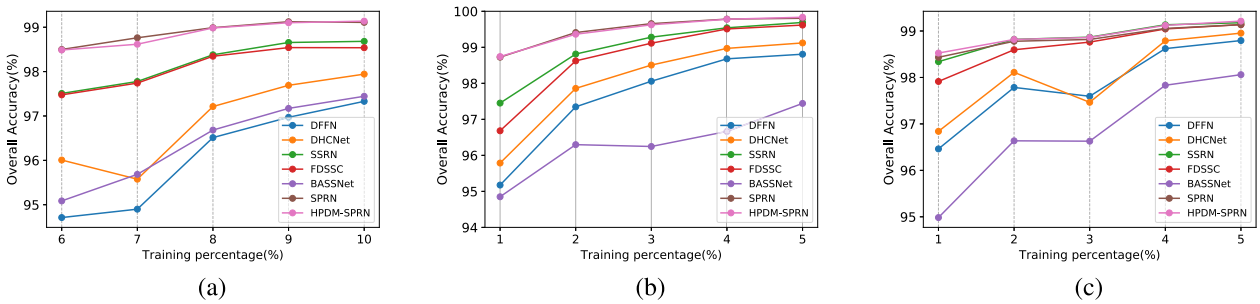


Fig. 11. Overall accuracy of different methods with different percentages of training samples on (a) Indian Pines, (b) Salinas, and (c) Pavia University.

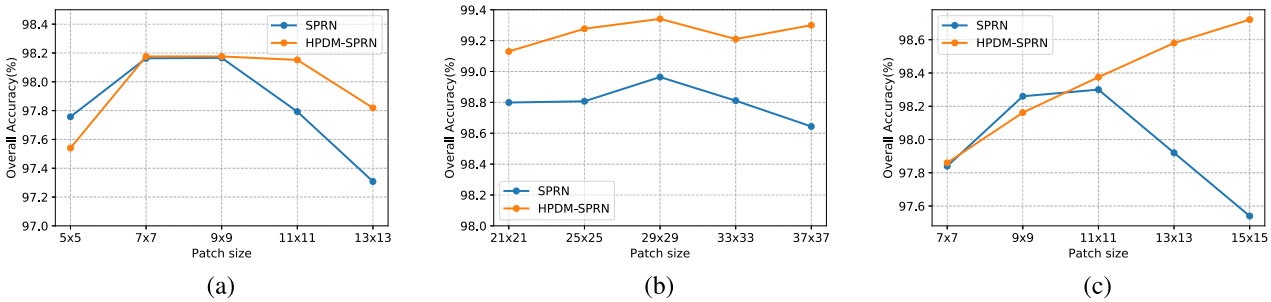


Fig. 12. Classification results of SPRN and HPDM-SPRN using different patch sizes on (a) Indian Pines dataset, (b) Salinas dataset, and (c) Pavia University dataset.

section, we will show this degradation phenomenon and verify our proposed HPDM aiming to capture the homogeneous pixels in a patch and alleviate the degradation of classification performance when using a large patch size. We first perform the experiments on SPRN. As illustrated in Fig. 12, there exists a peak when the patch size increases. The best patch size depends on different datasets and data sources. The best patch sizes on the Indian Pines are 7×7 and 29×29 on

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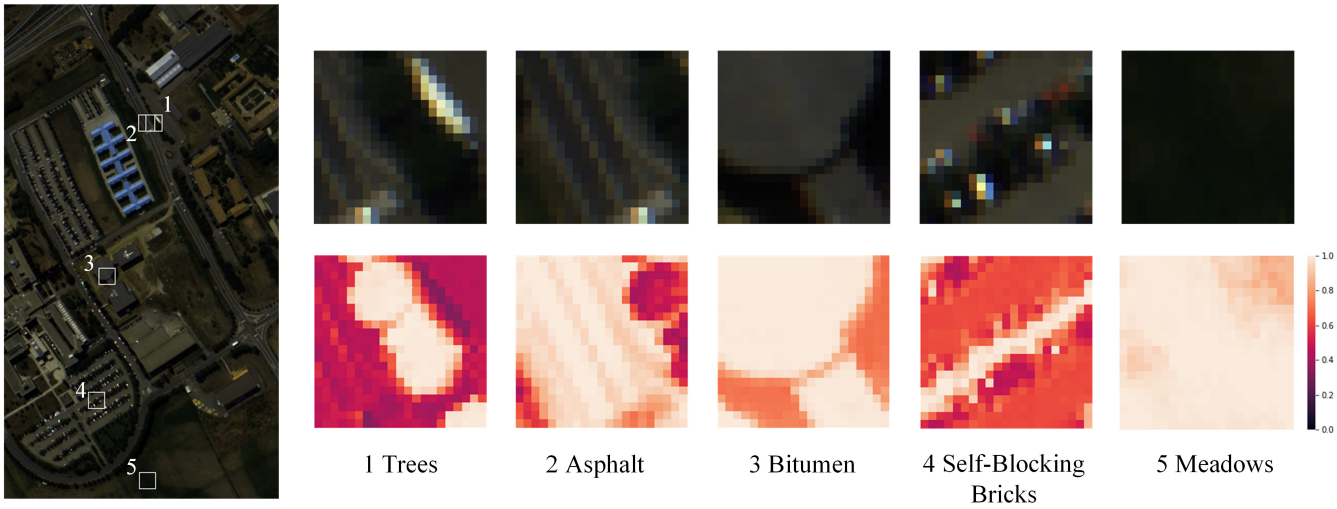


Fig. 13. Heatmaps of the weight masks of selected patches. (Left) False RGB imagery of Pavia University; the white frame is the selected patches. (Top row) (Right) Enlarged false RGB images. (Middle) Corresponding heatmaps and the bottom row are the corresponding labels. Light color means high weights, while the deep means low weights.

the Salinas and 11×11 on the Pavia University. When the patch size continues to increase, more heterogeneous pixels are included in the patch, which confuses the FE of CNN. Thus, the accuracies begin to decline. We add the HPDM to the head of SPRN, and the final results are showed as HPDM-SPRN. It can be observed that, as the patch size increases, the effect of the HPDM gradually reveals. The best patch size of HPDM-SPRN on the Indian Pines dataset is 9×9, and the degradation of accuracies is reduced as the patch size continues to increase. On the Salinas dataset, HPDM-SPRN gains higher accuracies than SPRN. On the Pavia University dataset, a larger patch size leads to better classification performance. Thus, by applying the spatial attention block, the choice of a patch size becomes less sensitive.

The heatmaps of weight masks can help us find out how the HPDM works. We train the HPDM-SPRN on the Pavia University dataset using the training samples illustrated in Tables II–IV and set the input patch size to 21×21 for a better illustration. Then, we select five patches and visualize their weight mask generated by HPDM in Fig. 13. In Fig. 13, the light color means high weight, and the deep means low weight. The label of the center pixel is the label of the patch. From the heatmap, we notice that the homogeneous pixels have relatively high values. Therefore, the influence of interfering pixels (with low weights) is reduced, and the final results benefit from a large patch size.

The HPDM can be integrated into other CNN-based HSI classification frameworks. We add the module in the head of networks mentioned in part D and perform the experiment on the Pavia University. The patch size is fixed to 21×21. Comparing Fig. 14 with Table VIII, some methods (SSRN, FDSSC, and SPRN) have a decrease in such large patch size. We also find that the HPDM works well in most frameworks. It alleviates the degradation phenomenon of SSRN, FDSSC, and SPRN and also helps to improve the accuracies of DFFN. However, the improvements on DHCNet and BASSNet are not significant. That is because DHCNet uses deformable convolutions, which are able to adaptively change the receptive field of convolution kernels. As for the BASSNet, the spatial size of the input patch is reduced to 1×1 after the first convolutions; thus, the spatial information may not be fully used.

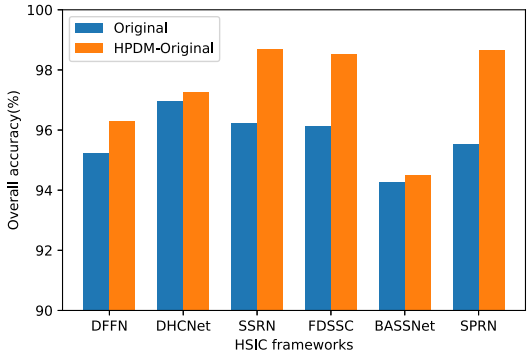


Fig. 14. Results of different frameworks by applying HPDM.

VI. Conclusion

In this article, an effective and efficient HSI classification framework, SPRN, has been proposed. The SP operation and parallel CNNs are implemented by using grouped convolutions. In order to obtain a better classification performance, we improve the residual block by adding an additional branch. According to the results on three popular datasets, our SPRN, lightweight and efficient, achieved state-of-the-art results, especially with limited training samples. In our method, the spectral bands are partitioned into groups with equal width. An adaptive partition strategy can be further explored.

In addition, we have proposed a novel spatial attention module named HPDM that can be integrated into any CNN-based HSI classification framework. The experiments show that the module is able to capture the homogeneous pixels in

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input patches and alleviate the degradation of classification performance as the input patch size increases.

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